Technical Documentation: Data Preprocessing & Cleaning

1. Introduction

This documentation provides the theoretical framework for the preprocessing pipeline implemented in the disease_risk prediction project. Data cleaning is a critical step in the machine learning workflow, ensuring that models learn from high-quality, representative data rather than noise or biases introduced by incomplete records.

2. Part A: Handling Missing Data

Missing data occurs when no value is stored for an observation in a variable. Leaving these empty can crash algorithms or bias results.

- **Simple Imputation (Mean/Mode):**
  - *Definition:* Replacing missing values with the average (mean) for numerical data or the most frequent value (mode) for categorical data.
- **KNN Imputation (K-Nearest Neighbors):**
  - *Definition:* A method that identifies the 'K' most similar records based on available features and uses their values to estimate the missing data point. It is effective for capturing local data patterns.
- **MICE (Multivariate Imputation by Chained Equations):**
  - *Definition:* An iterative approach where each missing variable is modeled as a function of all other variables. It creates multiple regression models to "chain" the predictions, making it one of the most robust imputation methods for complex datasets.

3. Part B: Outlier Detection & Treatment

Outliers are data points that differ significantly from other observations. They can skew statistical models, leading to poor predictive performance.

- **Z-Score Method:**
  - *Definition:* A statistical measure that describes a value's relationship to the mean of a group of values. A Z-score of >3 or <-3 indicates that the value is an extreme outlier (more than 3 standard deviations from the mean).
- **IQR (Interquartile Range) Method:**
  - *Definition:* Based on the range between the 25th (Q1) and 75th (Q3) percentiles. Values falling below Q1 - 1.5 * IQR or above Q3 + 1.5 * IQR are considered outliers.
- **Winsorization:**
  - *Definition:* A transformation that "caps" extreme values at a specific percentile (e.g., 1st and 99th). Unlike trimming (deleting rows), Winsorization retains the total number of records while limiting the influence of extreme values.

4. Part C: Final Dataset & Comparative Summary

After cleaning, it is essential to compare the "Raw" vs. "Cleaned" state.

- **Dataset Shape:** Ensuring the row count remains consistent (1,000 records) validates that the preservation strategy (capping vs. deletion) was successful.
- **Statistical Integrity:** Comparing the mean and standard deviation before and after cleaning ensures that the data's natural distribution was preserved throughout the process.

5. Conclusion

By applying these techniques, we successfully achieved a clean, complete, and balanced dataset. The use of **MICE imputation** preserved relationships between health variables, while **Winsorization** ensured that extreme health readings (like abnormally high BMI) remained in the dataset without causing the machine learning model to misinterpret them as standard cases.